7	**(a)**	minimum frequency of e.m. radiation/a photon (not "light")	M1	
		for emission of electrons from a surface	A1	[2]
		(*reference to light/UV rather than e.m. radiation, allow 1/2*)		

(b)		E_{MAX} corresponds to electron emitted from surface	B1	
		electron (below surface) requires energy to bring it to surface, so less than E_{MAX}	B1	[2]
(c)	**(i)**	$1/\lambda_0 = 1.85 \times 10^6$ *(allow 1.82 to 1.88)*	C1	
		$f_0 = c / \lambda_0$ $= 3.00 \times 10^8 \times 1.85 \times 10^6$ $= 5.55 \times 10^{14}$ Hz	A1	[2]
	(ii)	$\Phi = hf_0$		
		$= 6.63 \times 10^{-34} \times 5.55 \times 10^{14}$ *(allow ECF from **(c)(i)**)*	C1	
		$= 3.68 \times 10^{-19}$ J	A1	[2]
(d)		sketch: straight line with same gradient	M1	
		intercept between 1.0 and 1.5	A1	[2]